

Xiao Fu

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EDUCATION

Zhejiang University

Hangzhou, Zhejiang

Bachelor in Information Engineering | College of Information Science & Electronic Engineering Sept. 2018 – Jul. 2022

- **GPA:** 3.96/4.00, 90.54/100, **Ranking:** 4/145
- **National Scholarship** awardee, **Twice**
- **Chu Kochen Honors College:** Intensive Training Program of Innovation and Entrepreneurship (ITP) (Honored minor for the selected top 40 students with innovation capability from 6,000 freshmen)
- **Courses:** Computer Vision(96), Digital Signal Processing(95), Matrix Theory(94), Stochastic Process(96), Probability and Mathematical Statistics(95), Numerical Analysis(92), Signals and Systems(98), Calculus(A)(90)

PUBLICATIONS

* indicates equal contribution

- [1] Guangyi Zhang*, **Xiao Fu***, Qiyu Hu, Yunlong Cai, Guanding Yu, “Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network,” *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [[Paper](#)][[Code](#)][[Slides](#)]

RESEARCH EXPERIENCE

Faster Semantic/Instance Labeling with Panoptic NeRF

Max Planck Institute for Intelligent Systems

Supervisor: [Dr. Yiyi Liao](#) and [Prof. Andreas Geiger](#)

Jul. 2021 – Now

- Propose Panoptic NeRF, which transforms coarse 3D bounding box annotations (static objects in outdoor KITTI360 dataset) into dense and consistent 2D pixel-wise and 3D point-wise semantic and instance annotations
- Panoptic NeRF, which is based on NeRF, **leverages weak 2D semantic cues and weak 3D semantic supervision**. It takes DeepLab’s pretrained features as 2D priors to overcome the ambiguity of labeling in 3D overlapping areas and utilizes LiDAR, fisheye, and stereo cameras to improve the NeRF’s density
- Intend to publish an ECCV 2022 paper as the first author

Faster Solution of Optimization Algorithm with DLDUNN

Zhejiang University

Supervisor: [Prof. Guanding Yu](#)

Jun. 2020 – Feb. 2021

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network(DLDUNN) which unfolds the iterative Penalty Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award(Team leader) - Student Research Training Program(Province-level)**

HONORS AND SCHOLARSHIPS [[Certificate & Report](#)]

- National Scholarship (**Highest Honor for Chinese Undergraduates, Twice**) 2020,2021
- ISEE Honor Student Award (**Top 5 students among ISEE 314 undergraduates**) 2021
- ISEE Yuelun Alumni Scholarship (**Top 10 teams out of ISEE undergraduates and graduates**) 2020,2021
- First-Class Scholarship for Academic Excellence of Zhejiang University (**Top 3%**) 2020,2021
- Government Scholarship for Academic Excellence of Zhejiang Province (**Top 3%**) 2019
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 7% out of 26,112 teams**) 2021
- Gold Prize – 7th International “Internet+” Innovation and Entrepreneurship Contest (**Team Leader**) 2021
- Gold Prize – 12th “Challenge Cup” National College Student Business Plan Competition 2020
- Grand Prize – “Yongdian Cup” Innovation and Entrepreneurship Contest (**Team Leader**) 2019
- Grand Prize – “Midea” Global Technology Innovation Contest 2020
- Academic Excellence Scholarship of Zhejiang University 2019,2020,2021
- Zhejiang University Five-Star Volunteer 2021



浙江大学 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

CERTIFICATE

This is to certify that Mr Xiao Fu ranks the 4th among the 145 students with the specialty of Information Engineering.

College of Information Science & Electronic Engineering
Zhejiang University

